Integrating R with Power BI for Advanced Data Analytics

Introduction

Data visualization helps us understand data and identify useful patterns. In this practical, I worked on integrating R programming with Power BI to perform data analysis and create visualizations using the Sample Superstore dataset.

The dataset contains information about sales, profit, quantity, discounts, customers, and products from 2014 to 2017. The main objective was to clean the data, calculate summary statistics, create visualizations using R, and apply K-Means clustering for customer segmentation.

Setting Up R and Importing the Dataset

The first step was to install R and the ggplot2 package, which was used to create visualizations. I then configured the R installation in Power BI Desktop so that both tools could work together.

Next, I imported the Sample Superstore dataset into Power BI using the Text/CSV option. I used Power Query Editor to check the columns and assign the correct data types. For example, the order and shipping dates were converted to Date format, while sales, profit, and discount were set to Decimal Number.

Data Cleaning

Before performing the analysis, I checked the dataset for missing values, errors, and duplicate records. Using the column quality, column distribution, and column profile options in Power Query Editor, I found that the dataset had no missing values.

I also removed duplicate records by selecting all columns except Row ID. This reduced the dataset from 9,994 rows to 9,993 rows. Data cleaning was an important step because incorrect or repeated records could affect the results of the analysis.

Using R Scripts in Power BI

After cleaning the dataset, I used R scripts inside Power Query Editor to create two additional columns.

The first was Profit Margin %, which helped analyse profitability at the order-line level. The second was Discount Band, which grouped discounts into four categories. These columns made it easier to study how discounts were related to profitability.

I also created a separate query called Summary_Stats to calculate the mean, median, standard deviation, minimum, and maximum values for Sales, Quantity, Discount, and Profit. The results were displayed using a table visual in Power BI.

Creating Visualizations Using R

I created three R visuals in Power BI using the ggplot2 package. Each visualization helped analyse a different aspect of the dataset.

1. Distribution of Sales

I created a histogram to understand the distribution of sales values. Since the sales data was highly skewed, I used a logarithmic scale to make the distribution easier to interpret. This helped in understanding how order values were spread across the dataset.

2. Profit by Category

I used a boxplot to compare profit across Furniture, Office Supplies, and Technology. A boxplot is useful because it shows the spread of values and helps identify differences between categories. Extreme outliers were hidden to make the main distributions easier to read.

The visualization showed that profitability varied across categories, with Technology and Office Supplies being important contributors to overall profit.

3. Correlation Heatmap

I created a correlation heatmap using Sales, Quantity, Discount, and Profit. A heatmap represents relationships between numerical variables using colours and correlation values.

The results showed a positive relationship between Sales and Profit, while Discount and Profit had a negative relationship. This suggested that higher discounts could be associated with lower profits. However, correlation alone does not prove that discounts directly cause losses.

Advanced Analytics Using K-Means Clustering

The final part of the practical involved applying K-Means clustering for customer segmentation. Unlike ordinary charts, clustering helps group similar customers based on their purchasing behaviour.

I used total Sales, total Profit, and total Quantity for each customer as the clustering variables. These values were standardised using the scale() function so that variables with larger numerical ranges would not dominate the results.

I selected four clusters and used 25 random starts to improve the clustering process. A fixed random seed was also used to make the results reproducible.

The clusters were numbered according to their average sales, with Cluster 1 representing the lowest-spending customers. The customer segments were displayed using a scatter plot, where sales and profit helped show how customers differed from one another.

This analysis can help a business identify high-value customers, understand different purchasing patterns, and make better decisions about customer retention and marketing.

Building the Power BI Dashboard

Along with the R analysis, I created an interactive Power BI dashboard containing five visuals and two slicers. The dashboard included information about sales by region, sales trends, profit by category, and product performance.

The slicers allowed users to filter the data, while the drill-down feature made it possible to explore information at different levels of detail. Combining these features with R-based analytics made the dashboard more useful for exploring the dataset.

Key Observations

From the overall analysis, I observed that the business had a profit margin of 12.47% and sales growth of 20.4% in 2017. Technology and Office Supplies were important contributors to profit, and the West region performed well.

The analysis also highlighted discounting as an area that needed attention. Discounts above 20% and a group of 23 loss-making customers were identified as potential concerns for profitability. The results suggested that the business could benefit from reviewing deep discounts, especially for Furniture and Tables, and focusing on high-value customers.

Conclusion

This practical helped me understand how R can be integrated with Power BI to perform more advanced data analysis. I learned how to install and configure R, import and clean data, calculate summary statistics, create R visualizations, and apply K-Means clustering.

The practical also showed me that data visualization is not only about presenting numbers through charts. It can also help identify patterns, understand customer behaviour, and support business decisions. Overall, combining R with Power BI provided a useful way to move beyond basic dashboards and perform more detailed analysis of business data.